Mass Gap for the O(N) Model at Large N via Brascamp-Lieb and Feynman-Kac

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Abstract

We outline a proof of the mass gap for the O(N) linear sigma model in two Euclidean dimensions at large N. The proof has three steps: (1) the gap equation determines the saddle point σ_* and the physical mass $m_0 = \sqrt{\sigma_*}$ (dimensional transmutation), (2) Brascamp-Lieb gives σ-concentration at σ_* with $O(1/N)$ variance, and (3) Feynman-Kac controls the $1/N$ corrections to the propagator mass. The convexity condition for Brascamp-Lieb, $\sigma_* > 1/(16\pi\lambda)$, is a condition on the renormalized mass vs. the coupling and is cutoff-independent.

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1 Setup

1.1 The O(N) linear sigma model

Let $\varphi = (\varphi^1, \ldots, \varphi^N) : \Lambda \to \mathbb{R}^N$ on a lattice $\Lambda \subset \mathbb{Z}^2$. The action is

$$S[\varphi] = \tfrac{1}{2} \sum_i \langle \varphi^i, (-\Delta)\varphi^i \rangle + N \sum_x \lambda \big(\tfrac{|\varphi(x)|^2}{N} - v^2 \big)^2. \tag{1}$$

The bare parameter v^2 is adjusted as a function of the UV cutoff Λ to keep the physical mass fixed (the only renormalization needed in 2d; the quartic coupling λ does not run).

1.2 The σ-field: two definitions

There are two natural definitions of the σ-measure, related by a Laplace transform.

Definition A (pushforward). Define $s(x) = |\varphi(x)|^2/N$ as a measurable function of φ, and let μ_s be the pushforward of the interacting measure under s. By the disintegration theorem:

$$\langle F(\varphi) \rangle = \int \mathbb{E}[F(\varphi) \,|\, s_\varphi = s] \, d\mu_s(s). \tag{2}$$

Conditional on $s(x) = |\varphi(x)|^2/N$, each $\varphi(x)$ lives on a sphere $S^{N-1}_{\sqrt{Ns(x)}}$, coupled by the kinetic term. This is a well-defined positive measure, but the conditional expectation $\mathbb{E}[\varphi^i(x)\varphi^i(0) \,|\, s]$ is *not* exactly $(-\Delta + s)^{-1}(x, 0)$ — it is the propagator on the product of spheres, not on $\mathbb{R}^N$.

Definition B (HS auxiliary field). Use the Gaussian identity to linearize the quartic:

$$e^{-N\lambda(u-v^2)^2} = c \int d\sigma \; e^{-N\lambda\sigma^2 + 2N\lambda\sigma(u-v^2)}. \tag{3}$$

Applying this at each site with $u = |\varphi(x)|^2/N$, the φ-integral becomes Gaussian with covariance $(-\Delta + 4\lambda\sigma)^{-1}$. Integrating out φ exactly:

$$\langle \varphi^i(x)\varphi^i(0) \rangle = \mathbb{E}^{\mathrm{HS}}_\sigma \big[(-\Delta + 4\lambda\sigma)^{-1}(x, 0) \big] \tag{4}$$

where $\mu^{\mathrm{HS}}_\sigma \propto \det(-\Delta + 4\lambda\sigma)^{-N/2} e^{-N\sum_x [\lambda\sigma^2 - 2\lambda\sigma v^2]} \, d\sigma$. This is **exact** at every N: the σ-integral reproduces the original φ-integral. But σ here is an auxiliary integration variable, not literally $|\varphi|^2/N$.

Relation: Laplace transform. The two definitions are connected by the identity

$$\mathbb{E}_{\mathrm{GFF}} \big[e^{-\alpha \cdot s} \big] = \frac{\det(-\Delta + 2\alpha/N)^{-N/2}}{\det(-\Delta)^{-N/2}}, \tag{5}$$

which expresses $\det(-\Delta+\tau)^{-N/2}$ (from the HS integral) as the moment generating function of the pushforward $s(x) = |\varphi(x)|^2/N$ under the free GFF, with $\tau = 2\alpha/N$. At $N = \infty$ (saddle-point of the Laplace transform), the two definitions agree: $s_* = \sigma_* = \langle|\varphi|^2/N\rangle$ and both produce the propagator $(-\Delta+\sigma_*)^{-1}$. At finite N, they differ by $O(1/N)$ corrections from the saddle-point approximation of the inverse Laplace transform.

Convention. For the proof, we use Definition B (HS auxiliary field), which gives the exact formula (2). Rescaling $\tau = 4\lambda\sigma$ and absorbing constants into s_{eff}:

$$s_{\text{eff}}[\tau] = \tfrac{1}{2}\mathrm{Tr}\log(-\Delta+\tau) + \sum_x V(\tau(x)), \tag{6}$$

where $V(\tau) = \tau^2/(16\lambda) - \tau v^2/2 + \text{const}$, and (2) becomes $\langle\varphi^i\varphi^i\rangle = \mathbb{E}_\tau[(-\Delta+\tau)^{-1}]$.

2 The gap equation

2.1 Saddle point of s_{eff}

For uniform $\sigma(x) = \sigma_0$, the stationarity condition $ds_{\text{eff}}/d\sigma_0 = 0$ is

$$\tfrac{1}{2}G(x,x;\sigma_*) = 2\lambda(v^2-\sigma_*), \tag{7}$$

where $G(x,x;\sigma_0) = \int \frac{d^2k}{(2\pi)^2}\frac{1}{k^2+\sigma_0}$. Note that v^2 is *not* a stationary point: at $\sigma_0 = v^2$ the derivative $\frac{1}{2}G(x,x;v^2) > 0$, so the minimum lies at $\sigma_* < v^2$.

In 2d with cutoff Λ: $G(x,x;\sigma_*) \approx \frac{1}{4\pi}\log\frac{\Lambda^2}{\sigma_*}$, giving

$$\frac{1}{8\pi}\log\frac{\Lambda^2}{\sigma_*} = 2\lambda(v^2-\sigma_*). \tag{8}$$

For $\sigma_* \ll v^2$: $\sigma_* \approx \Lambda^2 e^{-16\pi\lambda v^2}$, exponentially small (dimensional transmutation).

2.2 Physical mass and renormalization

The physical mass at $N = \infty$ is $m_0 = \sqrt{\sigma_*}$, since $\sigma = \sigma_*$ deterministically and $(-\Delta+\sigma_*)^{-1}(x,0) \sim e^{-m_0|x|}$.

In the continuum limit ($\Lambda \to \infty$), we adjust the bare parameter:

$$v^2(\Lambda) = \sigma_* + \frac{1}{16\pi\lambda}\log\frac{\Lambda^2}{\sigma_*} \tag{9}$$

to keep σ_* (and hence m_0) fixed. The quartic coupling λ does not run (super-renormalizable in 2d).

3 Brascamp-Lieb concentration

3.1 Hessian at σ_*

The Hessian of s_{eff} at σ_*, in Fourier, is

$$H(k) = 2\lambda - \tfrac{1}{2}\Pi(k), \quad \Pi(k) = \int \frac{d^2p}{(2\pi)^2} \frac{1}{(p^2+\sigma_*)((k-p)^2+\sigma_*)}. \tag{10}$$

At $k = 0$: $\Pi(0) = 1/(4\pi\sigma_*)$, so $H(0) = 2\lambda - 1/(8\pi\sigma_*)$.

3.2 Convexity condition

$H(k)$ is positive definite for all k when $H(0) > 0$, i.e.,

$$\boxed{\sigma_* > \frac{1}{16\pi\lambda}.} \tag{11}$$

This is a condition on the **renormalized** parameters σ_* and λ. It is **cutoff-independent**: since λ does not run and σ_* is held fixed by renormalization, the condition (8) either holds or fails independently of Λ. In particular, it survives the continuum limit.

Equivalently: the physical mass must satisfy $m_0 > 1/\sqrt{16\pi\lambda}$. For $\lambda = 1$: $m_0 > 0.14$.

3.3 Brascamp-Lieb bounds

When (8) holds, s_{eff} is strictly convex at σ_* with convexity parameter $\kappa = 2\lambda - 1/(8\pi\sigma_*) > 0$. The measure $e^{-Ns_{\text{eff}}}$ is log-concave, and Brascamp-Lieb [1] gives:

$$\text{Cov}(\sigma(x), \sigma(y)) \leq \frac{1}{N} H^{-1}_{xy}, \tag{12}$$

$$\text{Var}(\sigma(x)) \leq \frac{1}{\kappa N}, \tag{13}$$

$$|\kappa_k(\sigma(x))| \leq \frac{C^k\, k!}{(\kappa N)^{k-1}}, \quad k \geq 2. \tag{14}$$

The σ-field concentrates at σ_* with $O(1/\sqrt{N})$ fluctuations.

4 Feynman-Kac and $1/N$ corrections

4.1 Leading order ($N = \infty$)

At $N = \infty$: $\sigma = \sigma_*$ deterministically, the gap equation determines σ_*, and the mass is $m_0 = \sqrt{\sigma_*}$. No further analysis needed.

4.2 Finite N: cumulant expansion

From (2) and the Feynman-Kac formula, the φ-propagator involves $\mathbb{E}_\sigma[e^{-\int \sigma(\omega)\,dt}]$. The cumulant expansion:

$$\mathbb{E}_\sigma[e^X] = \exp\Big(\sum_{k=1}^{\infty} \frac{\kappa_k(X)}{k!}\Big), \quad X = -\int_0^s \sigma(\omega(t))\,dt. \tag{15}$$

- $\kappa_1 = -s\sigma_*$ (leading mass, from the gap equation).
- $\kappa_2 = O(s/(\kappa N))$ (from BL variance bound).
- $\kappa_k = O(s/(\kappa N)^{k-1})$ for $k \geq 3$ (from BL cumulant bounds).

The corrected mass:

$$m^2 = \sigma_*\big(1 + O(1/(\kappa N))\big). \tag{16}$$

For $N \gg 1/\kappa$: the correction is small and $m^2 > 0$.

4.3 Main theorem

Theorem 1 (Mass gap at large N)**.** *For the O(N) LSM on $\mathbb{Z}^2$ with $N \geq 3$ and $\sigma_* > 1/(16\pi\lambda)$ (convexity condition), there exists $N_0 = O(1/\kappa)$ such that for $N \geq N_0$, the connected two-point function decays exponentially:*

$$|\langle \varphi^i(x)\varphi^i(0)\rangle_c| \leq C\, e^{-m|x|} \tag{17}$$

with $m = \sqrt{\sigma_*}\,(1 + O(1/(\kappa N))) > 0$, where σ_* is the gap equation solution (4)*.*

Proof outline.

1. **Gap equation** (§2): σ_* is the minimum of s_{eff}, with $m_0 = \sqrt{\sigma_*} > 0$.
2. **Brascamp-Lieb** (§3): At σ_*, the Hessian is positive (by (8)), giving σ-concentration with $\text{Var} = O(1/(\kappa N))$ and cumulant bounds.
3. **Feynman-Kac** (§4): The cumulant expansion gives $m^2 = \sigma_*(1 + O(1/(\kappa N)))$. For $N > C/\kappa$: the correction is < 1 and $m > 0$.
4. **Exponential decay**: Standard from Feynman-Kac with $m > 0$. □

5 Discussion

5.1 The convexity condition is physical

The condition $\sigma_* > 1/(16\pi\lambda)$ is equivalent to $m_0^2 > 1/(16\pi\lambda)$, a lower bound on the physical mass. It is cutoff-independent and survives the continuum limit, since λ does not run in 2d and σ_* is the renormalized mass squared.

For strong coupling (λ large): the threshold $1/(16\pi\lambda) \to 0$, so the condition is easily satisfied. For any fixed $\lambda > 0$: the condition holds for $m_0 > 1/\sqrt{16\pi\lambda}$.

5.2 No cluster expansion needed

Unlike Kupiainen [2], who uses a cluster expansion to control the $1/N$ corrections for the NLSM, our proof uses only:

- The gap equation (a single self-consistent equation),
- Brascamp-Lieb concentration (a convexity-based inequality),
- Feynman-Kac cumulant expansion (path integral representation).

The key simplification: the convexity of s_{eff} at σ_* (when (8) holds) makes the σ-measure log-concave, replacing the cluster expansion with BL.

5.3 The $N = 2$ obstruction

For $N = 2$: the BKT transition occurs at finite λ (Dario-Garban [4]), driven by vortices ($\pi_1(S^1) = \mathbb{Z}$). The σ-field cannot detect angular topology. For $N \geq 3$: $\pi_1(S^{N-1}) = 0$, no BKT.

5.4 Comparison with Kupiainen

	This work	Kupiainen [2]
Model	LSM (finite λ)	NLSM ($\lambda = \infty$)
σ-control	Brascamp-Lieb	Cluster expansion
φ-decay	Feynman-Kac	Random walk expansion
Mass formula	$m = \sqrt{\sigma_*}(1 + O(1/N))$	$m = m_0(1 + O(1/N))$
Continuum limit	Yes (renormalized)	Yes

Since the LSM and NLSM share the same infrared physics (they differ only at scale $M_\sigma \sim \sqrt{\lambda}\, v$), both proofs establish the same physical mass gap in the large-N limit.

References

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